

Nurturing well-being through nature-based intervention: The mediating role of mindfulness state

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ABSTRACT

Hedonic well-being and eudaimonic well-being, respectively, emphasize maximizing pleasure and personal growth. While previous research has demonstrated that nature-based interventions enhance both forms of well-being, the underlying mechanisms remain unclear. Notably, mindfulness and nature-based interventions share overlapping benefits, including reduced fatigue and enhanced vitality, and thus, we hypothesize that mindfulness serves as a critical mediating mechanism through which nature-based interventions foster well-being. We also explore whether moments with higher states of mindfulness during nature-based interventions are associated with greater well-being outcomes. Study 1 employed the experience sampling method to measure happiness, stress, and autonomous motivation during intervention periods compared with those in baseline days, while Study 2 utilized the day reconstruction method to assess happiness and nature connectedness. Multilevel regression analyses revealed that, compared to baseline days, nature-based interventions increased happiness and decreased stress while enhancing autonomous motivation and nature connectedness. Importantly, mindfulness states mediated the relationship between nature-based intervention and well-being outcomes. Furthermore, during the nature-based intervention, activities with moments of heightened mindfulness were associated with enhanced well-being. This research highlights mindfulness states as a mediating mechanism in nature-based interventions and suggests that incorporating mindfulness elements into future nature-based intervention designs can maximize their beneficial impact on well-being.

1. Engagement with natural environments improves well-being

Well-being can be broadly categorized into hedonic and eudaimonic dimensions. Hedonic well-being emphasizes the maximization of pleasure and the minimization of pain, whereas eudaimonic well-being highlights the pursuit of meaning beyond pleasure-driven happiness, including being true to oneself and striving for personal growth (Disabato et al., 2016).

Nature-based interventions have been designed as activities or programs aimed at engaging individuals in nature-based experiences to enhance their well-being (Shanahan et al., 2019). Regarding hedonic well-being, meta-analyses demonstrate that across various contexts and cultures, contact with nature enhances positive emotions and alleviates negative ones (McMahan & Estes, 2015). Even a single exposure to

nature can substantially elevate positive emotional states (White et al., 2017). Research further indicates that awe-inspiring natural landscapes can effectively diminish stress responses (Bai et al., 2017). Corazon et al. (2019) reviewed the literature on the effectiveness of nature-based interventions from 2010 to 2019 and found that these interventions reduce stress while promoting positive emotions. On the other hand, previous literature on nature-based interventions for eudaimonic well-being has predominantly focused on improvements in interpersonal aspects, including reductions in social isolation (Maund et al., 2019) and enhancements in interpersonal relationships (Korpela et al., 2016). Additional dimensions of eudaimonic well-being, such as autonomous motivation and nature connectedness, warrant further exploration in the context of nature-based interventions.

Autonomous motivation serves as a crucial indicator of eudaimonic

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well-being (Ryan & Deci, 2000). Yang et al. (2022) found that participants who interacted with nature, compared to those in urban environments, reported greater feelings of autonomy, competence, and relatedness—the three core psychological needs that foster the development of autonomous motivation. Nature connectedness represents another important indicator of eudaimonic well-being. Wilson's (1984) biophilia hypothesis suggests that eudaimonic well-being fundamentally depends on one's relationship with nature. Consistent with this perspective, Mayer and Frantz (2004) introduced the concept of nature connectedness, defining it as individuals' emotional connection to the natural world. Unlike the emphasis on personal motivational states, nature connectedness highlights the inclusion of the natural environment within the boundaries of the self, thereby fostering a sense of resonance with nature (Kals et al., 1999). However, excessive self-focus may hinder their connection with the environment and potentially diminish their well-being (Frantz et al., 2005). Therefore, incorporating nature connectedness as an indicator provides meaningful insight into dimensions of eudaimonic well-being, especially regarding the integration between self and the natural world. The present study aims to examine the effects of nature-based interventions on eudaimonic well-being through the lens of both autonomous motivation and nature connectedness.

2. Mindfulness may mediate the effects of nature-based interventions on well-being

Mindfulness is defined as “the awareness that emerges through paying attention on purpose, in the present moment, and non-judgmentally to the unfolding of experience moment by moment” (Kabat-Zinn, 2003, p. 145). The mechanisms of mindfulness and the effects of nature on reducing fatigue have considerable overlap (Vitaglano et al., 2023). Ulrich et al. (1991) proposed the psycho-physiological stress recovery theory, suggesting that interactions with natural environments can reduce sympathetic nervous system activation, thereby lowering fatigue levels. Additionally, Kaplan (1995) introduced the Attention Restoration Theory (ART), suggesting that mental fatigue can be alleviated through direct exposure to natural stimuli, which helps restore depleted attention. According to ART, soft, fascinating natural stimuli are ideal for drawing people away from mentally exhausting environments, enabling them to feel connected to larger purposes and recover from cognitive fatigue, thereby regaining focus on their goals (Kaplan, 1995; Ohly et al., 2016). Previous studies also found that mindfulness training could also reduce burnout during work processes (van der Meulen et al., 2021; Luken & Sammons, 2016).

Enabling people to experience vitality is another overlapping mechanism of nature exposure and mindfulness states. Vitality is defined as a type of physical or mental energy that makes people feel their own aliveness, enthusiasm, and energy available to themselves (Ryan & Frederick, 1997). Past research shows that, compared to indoor activities, engaging in natural environments leads to greater subjective vitality (Ryan et al., 2010). Similarly, a mindfulness state has shown a positive correlation with subjective vitality (Brown & Ryan, 2003; Visser et al., 2015).

Given that mindfulness and nature-based interventions share overlapping benefits, including reduced fatigue and enhanced vitality, mindfulness states may be an important mediating variable influencing the effects of nature-based interventions on both hedonic and eudaimonic well-being.

Recently, a new form of mindfulness training, termed nature-based mindfulness, has emerged. Nature-based mindfulness emphasizes conducting traditional mindfulness practices in natural environments to strengthen individuals' connection with nature and thereby enhance well-being (Burger et al., 2024). Previous meta-analytic research has demonstrated that nature-based mindfulness can achieve moderate effect sizes ($g = 0.56$) in improving overall well-being (Djermis et al., 2019). Some scholars have also explored the use of virtual reality to

simulate traditional mindfulness training in natural environments, discovering its effectiveness in reducing anxiety (Benchimol-Elkaim et al., 2024). Qualitative research has similarly found that five days of mindfulness training in natural environments can effectively alleviate stress and foster connections between self, others, and the environment (Djermis et al., 2023). However, these studies primarily emphasize how natural environments enhance the effectiveness of traditional mindfulness interventions, while giving limited attention to how interaction with nature-based activities itself may influence mindfulness states. This study aims to investigate whether mindfulness states can mediate the enhancing effects of nature-based interventions on hedonic and eudaimonic well-being.

3. The present study

The vast majority of nature-based interventions measure well-being by using questionnaires to compare differences in well-being before and after the intervention (Korpela et al., 2016; Maund et al., 2019). This approach captures only general reports of well-being during nature-based interventions, while overlooking moment-to-moment subjective experiences during specific activities in the nature-based intervention process (Kahneman et al., 2004). To address this limitation, the current study adopted ecological momentary assessment (such as the experience sampling method used in Study 1 and the day reconstruction method used in Study 2) to capture people's well-being experiences when they participated in nature-based intervention activities (Kahneman et al., 2004).

In this study, we capture hedonic well-being during the nature-based intervention process by measuring happiness and stress levels (Bai et al., 2017; McMahan & Estes, 2015; White et al., 2017), while assessing eudaimonic well-being through autonomous motivation and nature connectedness (Pritchard et al., 2020; Richardson et al., 2021; Yang et al., 2022). However, both the experience sampling method and the day reconstruction method represent intensive data collection approaches that impose significant response burdens on participants (Eisele et al., 2022). To reduce participant fatigue and minimize careless responses, we strategically measured happiness, stress, and autonomous motivation in Study 1, while measuring happiness and nature connectedness in Study 2, rather than simultaneously employing all measures across both studies.

Furthermore, since the mechanisms through which nature-based interventions bring about well-being are still unclear, this study specifically compares whether the differences in well-being between baseline days and days with nature-based intervention activities are mediated by mindfulness states. We hypothesize that mindfulness states mediate the relationship between nature-based interventions and improvements in well-being (hypothesis 1, see Fig. 1). Given that nature-based intervention encompasses various activities that elicit different levels of mindfulness across individuals, we also hypothesize that moments characterized by higher mindfulness states are associated with greater well-being during the intervention (hypothesis 2).

4. Study 1

Study 1 used experience sampling methods to capture participants' well-being and mindfulness state when engaging in nature activities compared to those on a regular day (baseline). The study aimed to observe whether nature-based interventions increased and consequently enhanced individual autonomous motivation, increased happiness, and reduced stress. Study 1 also examined whether higher levels of mindfulness state during nature-based intervention days had a greater impact on well-being.

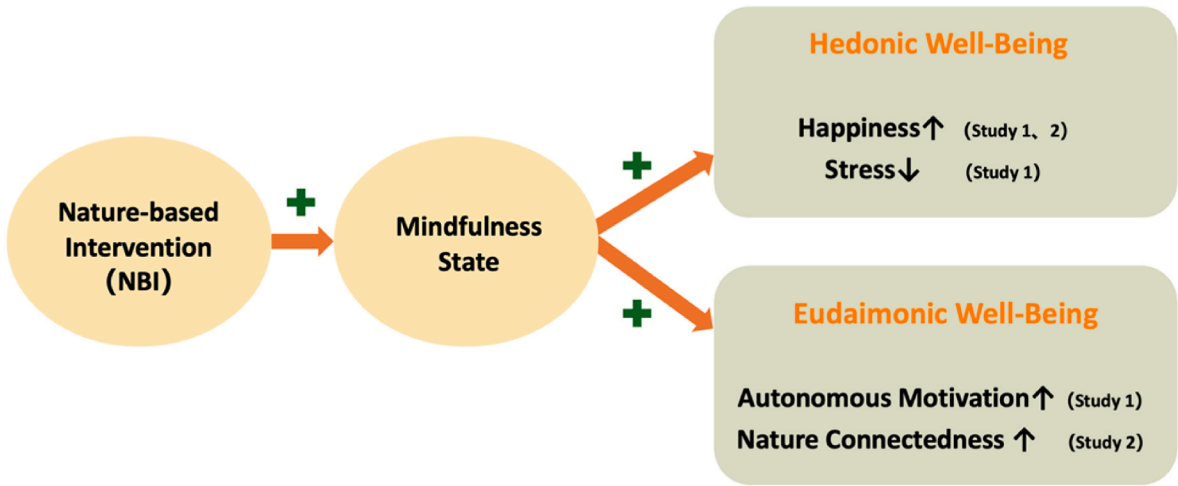


Fig. 1. Proposed mediation model of mindfulness states in the relationship between nature-based intervention and well-being.

4.1. Method

4.1.1. Nature-based interventions

The nature-based intervention in this research utilized the healing agricultural program developed by the Taiwan Leisure Agriculture Association (Yeh, 2024), the pioneer nature-based intervention combining local agricultural development and rural environments. This program was designed based on domesticated nature, aiming to enhance participants' well-being through participating in nature-based experiential activities. All activities in healing agricultural program were designed around four core elements (Fig. 2): Ecology, where participants left urban areas to immerse themselves in natural landscapes; Life, which involved interacting with farm animals and plants; Living, where

participants enjoyed seasonal local foods; and Production, where they sampled locally grown fruits and vegetables on the farm while learning about the growth processes of these products. Further details on the design elements of agricultural healing and the activity procedures in which participants engaged can be found in Supplemental Material A.

4.1.2. Procedure and participants

This study recruited participants online through various agriculture-related organizations. Therefore, participants were employees related to the agricultural industry (farmers, farm managers, etc.), and the participants themselves had some background in the agricultural industry. However, none of the participants had been exposed to nature-based intervention related courses, so participating agricultural activities to



Fig. 2. Four elements of the healing agricultural program.

enhance well-being was a completely new experience for them. Choosing agricultural industry employees as participants has one obvious advantage, as it allows us to capture the effects brought by the intervention itself, rather than simply the effects brought by the natural environment. Because they were exposed to natural environments on regular days, while on nature-based intervention days they were exposed to natural environments and simultaneously participated in interventions that interact with nature to enhance well-being, the comparison between the two can effectively capture the impact of the intervention activities themselves on well-being.

In this study, 13 participants (5 males, 8 females, mean age = 37.23, SD = 13.02) from Taiwan were recruited online during the summer of 2023. This study estimated the sample size based on the effect size of nature walking (versus urban walking) on positive emotions reported by Keenan et al. (2021). Given the longitudinal experimental design, the GLIMMPSSE software was used to estimate the necessary sample size for the planned multilevel regression analysis (Kreidler et al., 2013). The following parameters were used for the power analysis: Each participant completed at least 25 responses, with a within-subject variance of 0.7, and a scale factor of 0.5 for conservative effect size estimation. The results of this analysis indicated that a minimum of 10 participants would be required to detect the main effect of nature intervention on positive emotions with a statistical power of 0.8 and a Type I error rate of 0.05.

The study utilized an experience sampling method to assess participants' psychological processes during a nature-based intervention. Over 18 weeks, participants engaged in a 9-round cycle, alternating weekly between a full-day nature-based intervention session on Mondays (intervention days) and their regular routine on Mondays (baseline days). Participants attended a nature-based intervention session in the first week and returned to their regular routines in the second week. Each nature-based intervention session lasted for the whole day. Psychological indicators, including emotions, motivation, and mindfulness state, were measured on both the intervention days and baseline days. Following the experience sampling procedures outlined in Zeng and Yeh (2025), questionnaires were randomly sent three times per week via LINE messaging software, with a minimum 3-h interval between deliveries. Participants were required to complete each questionnaire within 30 min of receipt; otherwise, responses were not accepted. This research received ethical approval from the Research Ethics Committee of National Taiwan University (NTU-REC-202205HS011), and all participants provided informed consent prior to participation.

4.1.3. Measurement

The experience sampling questionnaires included several key dimensions: Emotional aspects: assessed through questions on current happiness ("How happy do you feel right now?") and stress levels ("How stressed do you feel right now?"); Motivational aspects: Before answering motivation questions, we first asked participants about their current activity, then had them answer motivation-related questions regarding that activity based on the Situational Motivation Scale developed by Guay et al. (2001). This included four items assessing intrinsic motivation ("I do this because I find the process interesting"), identified motivation ("I do this because it is good for me"), external motivation ("I do this because I should"), and amotivation ("I do this, but I'm not sure if it's worth doing"). Within the framework of Self-Determination Theory, intrinsic motivation and identified motivation belong to the category of autonomous motivation, while external motivation and amotivation fall under the category of controlled motivation (Ryan & Deci, 2000). The study by Blanke and Brose (2017) indicates that the two dimensions of mindfulness state—present focus and nonjudgmental—are important variables influencing people's well-being. Therefore, this study used a questionnaire to measure the present focus ("I am fully focused on the present moment.") and nonjudgmental ("I am non-judgmental about everything happening in the present moment."). Finally, the mindfulness state scores were calculated as the average of present focus and nonjudgmental scores. All

above items were rated on a 6-point Likert scale (1 = strongly disagree, 6 = strongly agree).

4.1.4. Data analysis

The RStudio software was used to perform multilevel regression analysis utilizing the lme4 package (version 1.1–34; Bates et al., 2015). The lme4 package relies on full information maximum likelihood estimation, which is particularly well-suited for handling missing data that occurs at random (Enders & Bandalos, 2001). Since the occasion variable was nested within the person variable, the intercept was set as a random effect, controlling for variations in intercepts across different participants. Furthermore, we also used a more complex random slope multilevel model to examine the model's performance after controlling for individual differences. The results indicated that the experience sampling analysis results were sufficiently stable and less susceptible to individual differences (see Supplemental Material B for details).

Additionally, each experience sampling item was centered around the person mean. For multilevel mediation analysis, the Mplus software was used, with 5000 bootstrap samples conducted to estimate confidence intervals for the indirect effects (Muthén & Muthén, 2017). When testing Hypothesis 1, Mondays with the nature-based intervention were coded as "1," and baseline days were coded as "0." The differences between the two conditions on mindfulness and well-being were compared, along with their mediation model. When testing Hypothesis 2, baseline day data were excluded to observe the impact of mindfulness states during the nature-based intervention process on well-being. All raw data and codes were stored in the OSF platform (https://osf.io/28yt5/?view_only=a18373edcf9b48f1845a567217ea8618).

4.2. Results

A total of 741 experience sampling questionnaires were distributed throughout the study, with participants responding 298 times, resulting in an overall response rate of 40.2 %. Based on the findings of van Berkel (2019), experience sampling schedules designed with random signal prompts tend to induce significant response fatigue among participants, so a response rate of around 50 % was considered normal.

This study observed a significant decline in questionnaire response rates over the course of the intervention (Estimate = -4.38 , $p < .001$), suggesting participant fatigue from repeated questionnaire completion led to missing data. The time of day (morning, afternoon, or evening) had no significant effect on response rates ($X^2(2) = 4.71$, $p = .009$). Each completed experience sampling questionnaire was recorded as one activity token. A total of 199 activity tokens were collected on Mondays during nature-based intervention sessions, compared to 99 tokens on baseline days.

This study investigated whether emotions, motivation, and mindfulness states differed between Mondays with nature-based interventions and baseline days. The results revealed that (Table 1), compared to baseline days, participants experienced higher happiness on the days of the nature-based intervention (Estimation = 0.63 , $p < .001$) and less stress (Estimation = -0.52 , $p = .002$). Compared to baseline days, participants reported more intrinsic motivation (Estimation = 0.66 , $p < .001$), identified motivation (Estimation = 0.33 , $p < .001$), and external motivation (Estimation = 0.23 , $p = .004$) on intervention days, but there was no significant difference in amotivation (Estimation = -0.14 , $p = .263$). Furthermore, compared to baseline days, participants reported higher mindfulness state (Estimation = 0.5 , $p < .001$).

Additionally, we sought to examine whether the state of mindfulness could explain the differences in well-being outcomes between nature-based interventions and baseline days. The results from the within-subject mediation model showed that nature-based intervention, compared to baseline days, enhanced participants' intrinsic motivation, which in turn increased their happiness (Indirect Effects, 95 % CI [0.076, 0.608]) and reduced their stress levels (Indirect Effects, 95 % CI

Table 1

Comparison of emotions, motivation, and mindfulness states between nature-based interventions and baseline days.

	Nature based Intervention		Baseline		Estimation	p
	M	SD	M	SD		
Emotion						
Happiness	4.85	1.05	4.41	1.10	0.63	<0.001***
Stress	2.56	1.45	2.86	1.48	-0.52	0.002**
Activity motivation						
Intrinsic motivation	4.91	0.93	4.39	1.14	0.66	<0.001***
Identified motivation	4.99	0.85	4.83	0.85	0.33	<0.001***
External motivation	4.97	0.86	4.91	0.78	0.23	0.004**
Amotivation	2.43	1.21	2.53	1.2	-0.14	0.263
Mindfulness State	4.69	0.99	4.34	1.04	0.5	<0.001***

Note. Mondays with the nature-based interventions were coded as "1," and baseline days were coded as "0." The estimation represents the effect of comparing the differences between the two using multilevel regression analysis, with intercepts treated as random effects. $n = 298$ observations from 13 persons. ** $p < .01$; *** $p < .001$.

Table 2

Within-subject mediation model of the impact of the nature-based intervention on well-being.

	Indirect Effects 95% Confident Intervals
NBI→Mindful→Happiness	[0.035, 0.355]
NBI→Mindful→Stress	[-0.288, -0.026]
NBI→Mindful→Intrinsic	[0.034, 0.358]
NBI→Mindful→Identified	[0.039, 0.298]
NBI→Mindful→External	[0.041, 0.289]
NBI→Mindful→Amotivation	[-0.451, -0.055]

Note. NBI = Nature-based intervention. For the NBI variable, Mondays with the nature-based intervention were coded as "1," and baseline days were coded as "0." The indirect effects were calculated with a 95 % confidence interval after 5000 bootstrap iterations, and when the confidence interval did not include 0, it indicated that the mediation effect is significant. Thus, each mediation effect in this table met the significance criteria. $n = 298$ observations from 13 persons.

[-0.746, -0.110]). Furthermore, compared to baseline days, nature-based interventions effectively promote mindfulness states, which in turn facilitate increases in intrinsic motivation (Indirect Effects, 95 % CI [0.034, 0.358]), identified motivation (Indirect Effects, 95 % CI [0.039, 0.298]), and external motivation (Indirect Effects, 95 % CI [0.041, 0.289]), while reducing amotivation (Indirect Effects, 95 % CI [-0.451, -0.055]) (Table 2).

Furthermore, we also aimed to understand the mindfulness state behind participants' engagement in various activities throughout the entire day of the nature-based intervention and how the mindfulness

state influenced emotional experiences and motivation. Note that our focus here was on all activities during the nature-based intervention day to compare different motivations and emotional outcomes. Individuals varied in their levels of mindful engagement across all nature-based activities. For example, some experienced the highest state of mindfulness while enjoying delicious food, whereas others felt most mindful when interacting with animals. Therefore, we did not specifically analyze the types of activities but focused instead on the varying levels of mindfulness experienced during participation in nature-based interventions. The results from all activities showed that moments with higher mindfulness state were associated with greater feelings of happiness, fewer feelings of stress, more intrinsic motivation, more identified motivation, more external motivation, and less amotivation (Table 3).

4.3. Discussion

Our results revealed that, compared to baseline days, participants reported significantly higher happiness and lower stress (hedonic well-being), as well as greater intrinsic and identified motivation (eudaimonic well-being), on nature-based intervention days. Although this study found that nature-based intervention had a promoting effect on autonomous motivation, the unexpected finding that nature-based intervention also promoted external motivation did not align with previous research, as Yang et al. (2022) suggested that nature boosted psychological needs that, per self-determination theory, were inversely related to external motivation. This may have been because participants were assigned some homework tasks after each nature-based intervention session, to better enable participants to reflect on or savor the content of that week. However, this homework design had potentially eliminated participants' autonomy and forced participants to seriously

Table 3

The impact of activity evaluation on well-being during the nature-based intervention.

	Estimation	Confident Interval	p
Using mindfulness to predict:			
Happiness	0.49	4.35 – 5.12	<0.001***
Stress	-0.47	-0.72 – -0.22	<0.001***
Intrinsic motivation	0.28	0.16 – 0.40	<0.001***
Identified motivation	0.28	0.17 – 0.39	<0.001***
External motivation	0.23	0.12 – 0.34	<0.001***
Amotivation	-0.39	-0.58 – -0.20	<0.001***

Note. This table presents the experience sampling questionnaire results for the days of the nature-based intervention, excluding data from baseline days. The estimates represent the effects derived from multilevel regression analyses with random intercepts. $n = 199$ observations from 11 persons during intervention days. *** $p < .001$.

engage in each nature-based intervention session due to external assignment requirements.

Study 1 results support hypothesis 1. Further mediation analysis revealed that nature-based intervention days yielded higher mindfulness states than baseline days, which in turn enhanced happiness, intrinsic motivation, and identified motivation, as well as reduced stress. The results also support hypothesis 2. When we focus only on the nature-based intervention day, results showed that moments with higher mindfulness states were associated with greater well-being outcomes.

5. Study 2

Study 1 utilized the experience sampling method, which could only randomly capture three activities per day, making it difficult to record the full scope of activities throughout the day during the nature-based intervention. To address this, Study 2 employed the day reconstruction method, allowing participants to report all activities from both the nature-based intervention and baseline days and the psychological process associated with each activity (Kahneman et al., 2004). This method enabled more comprehensive data collection from a smaller sample and provided a broader perspective on effectively designing nature-based interventions. In addition, since Study 1 collected questionnaires at random time points, it placed a greater burden on participants (van Berkel et al., 2019), ultimately resulting in only 298 data points being collected. To address this problem, Study 2 employed the day reconstruction method with fixed time points for questionnaire collection, reducing the response pressure on participants and consequently gathering more activities for analysis.

Study 2 examined nature connectedness as a key indicator of eudaimonic well-being. McMahan et al. (2018) found that well-being improvements depend on both exposure to nature and a meaningful sense of connection to the natural environment, with mere exposure to natural settings insufficient without fostering this connection. Given that participants with agricultural backgrounds regularly encountered natural environments on baseline days, the critical factor in nature-based interventions was cultivating and strengthening nature connectedness through intervention activities, which is essential for enhancing well-being.

Meredith et al. (2020) highlighted that the duration of time spent in natural activities significantly influences the effectiveness of nature-based interventions. Additionally, as nature-based interventions often involve teamwork, interpersonal interactions may influence mindfulness states (Lindsay et al., 2019). Thus, Study 2 included activity duration and interpersonal interaction as control variables to examine whether the beneficial effects of nature-based interventions on well-being remain robust after controlling for these factors.

5.1. Method

5.1.1. Procedure and participants

Study 2 followed the same participant recruitment and intervention procedures as Study 1. In the summer of 2024, 11 Taiwanese participants (9 females, 2 males; Mean age = 34.91 years, SD = 11.18) were recruited. Based on the sample size estimation from Study 1, this sample was deemed sufficient. Similar with Study 1, participants completed day reconstruction questionnaires on Mondays during nature-based intervention weeks and on baseline days when returning to their regular routines. This cycle was repeated six times over 12 weeks.

On both baseline days and intervention days, participants were required to report every activity from the time they woke up until noon in a midday report at 12 p.m., followed by an afternoon report at 5 p.m. covering activities from noon to 5 p.m., and an evening report at 9 p.m. covering activities from 5 p.m. to 9 p.m. Participants were instructed to complete each report within 1 h of receiving the questionnaire, after which the opportunity to submit their responses would expire.

5.1.2. Measurement

In the day reconstruction questionnaire, participants first reported the activities they engaged in throughout the day, with a minimum of 3 activities and a maximum of 10 activities to be filled in. For each reported activity, participants were asked to report how long each activity lasted (counted in 15-min units, reporting how many 15-min intervals the activity lasted), and were then assessed on various psychological dimensions: Emotional aspect: Assessed through questions on happiness ("I feel happy during the activity"); Nature aspect: Measured by questions on nature connectedness ("I feel connected to nature during the activity"). Mindfulness aspect: Measured by questions on present focus ("I maintain focus on the present moment during the activity") and non-judgmental attitude ("I am non-judgmental about what was happening during the activity"). Finally, participants were asked to respond to an interpersonal-related item ("I interacted and collaborated with others during the activity"). Each question was answered using a 6-point Likert scale (1 = strongly disagree, 6 = strongly agree). In summary, at each sampling time point, participants completed a minimum of 18 questions and a maximum of 60 questions to capture psychological assessments of specific time period activities.

5.1.3. Data analysis

The analytical approach in this study was similar to that of Study 1, utilizing multilevel regression and multilevel mediation analysis to process the data. The preliminary analysis compared whether there were differences in duration and interpersonal interactions between events occurring in daily life and nature-based intervention activities. Results found that the duration of each event during nature-based interventions was significantly shorter than on baseline days (Estimation = -1.15 , $p < .001$). However, there was no difference in interpersonal interactions between nature-based interventions and baseline days (Estimation = 0.06 , $p = .425$). Therefore, in subsequent multilevel regression analyses, only duration was included as a control variable, rather than interpersonal interactions. Study 2 also used a more complex random slope multilevel model to examine the model's performance after controlling for individual differences. The results showed some instability (see Supplemental Material B for details), indicating that individual differences are an important factor influencing the reconstruction method. This may be because the day reconstruction method collected all daily activity data from each participant's daily life, thereby increasing the variability in their data and resulting in greater individual differences.

5.2. Results

In this study, a total of 396 survey samples were distributed, with participants completing 210 of them, resulting in an overall response rate of 53 %. This study also found a trend of declining questionnaire response rates as the course progressed (Estimation = -8.51 , $p = .016$), and this result was similar to the participant fatigue issue depicted in Study 1. Additionally, consistent with Study 1, time of day (morning, afternoon, or evening) did not affect participants' response rates ($X^2(2) = 0.8$, $p = .61$). Participants reported a total of 912 activity tokens, with 506 activity tokens occurring on Mondays during the nature-based intervention and 406 activity tokens occurring on baseline days.

Study 2 compared the differences in emotions, the degree of nature connectedness, and the mindfulness state between days with a nature-based intervention and baseline days. Results showed that on nature-based intervention days (Table 4), after controlling for the influence of activity duration, participants reported higher levels of happiness compared to baseline days (Estimation = 0.12 , $p = .032$), stronger nature connectedness (Estimation = 0.38 , $p < .001$), and greater mindfulness state (Estimation = 0.28 , $p = .002$).

Additionally, the within-subject mediation model in Table 5 indicated that mindfulness states could explain the increase in happiness and nature connectedness observed during the nature-based intervention compared to baseline days. The results found that the enhanced

Table 4

Comparison of emotions, nature connectedness, and mindfulness states between nature-based interventions and baseline days.

	Nature based Intervention		Baseline		Estimation	<i>p</i>
	M	SD	M	SD		
Happiness	5.11	0.93	4.94	1.00	0.12	0.032*
Nature Connectedness	4.79	1.49	4.37	1.43	0.38	<0.001***
Mindfulness State	5.16	0.77	4.99	0.89	0.28	0.002**

Note. Mondays with the nature-based intervention were coded as "1," and baseline days were coded as "0." The estimation represents the effect of comparing the differences between the two using multilevel regression analysis, with intercepts treated as random effects. All regression results were obtained after controlling for the influence of activity duration. *n* = 912 observations from 11 persons. **p* < .05; ***p* < .01; ****p* < .001.

Table 5

Within-subject mediation model of the impact of nature-based intervention on well-being.

	Indirect Effects
	95% Confident Intervals
NBI→Mindful→Happiness	[0.005, 0.324]
NBI→Mindful→Nature Connectedness	[0.020, 0.100]

Note. NBI = Nature-based Intervention. For the NBI variable, Mondays with the nature-based intervention were coded as "1," and baseline days were coded as "0." The indirect effects were calculated with a 95 % confidence interval after 5000 bootstrap iterations, and when the confidence interval did not include 0, it indicated that the mediation effect was significant. Conversely, if the confidence interval included 0, the mediation effect was not significant. All mediation effects were obtained after controlling for the influence of activity duration. *n* = 912 observations from 11 persons.

Table 6

The impact of activity evaluation on emotions during nature-based intervention.

	Estimation	Confident Interval	<i>p</i>
Using mindfulness to predict:			
Happiness	0.35	0.30 – 0.39	<0.001***
Nature Connectedness	0.32	0.23 – 0.42	<0.001***

Note. This table presents the day reconstruction questionnaire results for the days of the nature-based intervention, excluding data from baseline days. The estimates represent the effects derived from multilevel regression analyses with random intercepts. All regression results were obtained after controlling for the influence of activity duration. *n* = 506 observations from 11 persons during intervention days. ****p* < .001.

happiness associated with nature-based intervention, compared to baseline days, was primarily driven by greater mindfulness states (Indirect Effects, 95 % CI [0.005, 0.324]). In addition, mindfulness states were also a mediating variable through which nature-based interventions (compared to baseline days) brought about nature connectedness (Indirect Effects, 95 % CI [0.020, 0.100]).

This study focused on the activities occurring on the nature-based intervention days, excluding baseline days' activities, to explore which types of activities contribute most to increased happiness. Following the analytical approach of Study 1, we focused solely on the impact of mindfulness during the activity on well-being, as individuals differed in their levels of mindful engagement even when participating in the same activity. The results revealed that, after controlling for the influence of activity duration, activities that foster a greater mindfulness state were associated with higher levels of happiness (Estimation = 0.35, *p* < .001) and nature connectedness (Estimation = 0.32, *p* < .001) (Table 6).

5.3. Discussion

Study 2 found that nature-based interventions led to greater happiness and nature connectedness compared to baseline days. Furthermore, the results of Study 2 also showed that mindfulness states served as the mediating mechanism through which nature-based interventions, compared to baseline days, led to greater happiness and nature

connectedness, supporting hypothesis 1. Additionally, the study revealed that incorporating mindfulness state during the nature-based intervention was associated with enhanced well-being outcomes (participants' happiness and nature connectedness), supporting hypothesis 2.

6. General discussion

6.1. Nature-based interventions can enhance both hedonic well-being and eudaimonic well-being

Study 1 and Study 2 of this research were conducted at different times, with different participant groups, and employed distinct methodologies (experience sampling method and day reconstruction method) to examine the effectiveness of same nature-based intervention program. After integrating the results of Study 1 and Study 2, we found that, compared to baseline days, nature-based interventions enhanced hedonic well-being (i.e., increased happiness and reduced stress) and eudaimonic well-being (i.e., increased autonomous motivation and nature connectedness). In terms of hedonic well-being, these research results are consistent with previous findings on nature-based interventions enhancing positive emotions (Corazon et al., 2019; Maund et al., 2019). The results of this study also confirm the hypothesis mentioned in the psycho-physiological stress recovery theory that contact with nature can

reduce stress (Ulrich et al., 1991). As for eudaimonic well-being, unlike previous research that explored the effects of nature-based interventions on interpersonal aspects (Korpela et al., 2016; Maund et al., 2019), this study found that nature-based interventions can also promote psychological well-being dimensions such as autonomous motivation and nature connectedness.

Importantly, while previous research has relied on pre-post testing methods to assess the overall well-being effects of nature-based interventions (Korpela et al., 2016; Maund et al., 2019), this study employed ecological momentary assessment to capture real-time (Study 1) and recent-time (Study 2) well-being experiences during activities. This approach yielded findings consistent with prior research while providing enhanced ecological validity for understanding the beneficial effects of nature-based interventions on well-being.

6.2. Mindfulness state mediates the relationship between nature-based interventions and enhanced well-being

The results of both Study 1 and Study 2 revealed that mindfulness states were the mediating mechanism explaining why structured nature-based interventions provide greater well-being than daily life activities, supporting Hypothesis 1. Prior research demonstrated that passive viewing of nature videos enhanced mindfulness states and psychological well-being compared to urban videos (Stewart & Haaga, 2018). However, such video-based approaches lack essential bodily sensations—including improved air quality and tactile ground contact—that are integral to mindful experience (Khouri et al., 2017). This study employed actual nature-based interventions to more comprehensively examine the mediating role of mindfulness states in the relationship between genuine nature engagement and enhanced well-being. According to Kaplan's (1995) Attention Restoration Theory and Ryan et al.'s (2010) vitality theory, active and intentional engagement with natural environments effectively alleviates attentional fatigue and promotes feelings of vitality. Correspondingly, research on mindfulness interventions has demonstrated that 'acting with awareness' can significantly reduce fatigue among employees experiencing burnout (van der Meulen et al., 2021; Luken & Sammons, 2016). Additionally, previous studies have found positive correlations between mindfulness state and subjective vitality (Brown & Ryan, 2003; Visser et al., 2015). These theoretical and empirical parallels suggest overlapping mechanisms between mindfulness states and established nature-restoration theories, providing theoretical support for mindfulness as a mediating mechanism in enhancing well-being through structured nature intervention.

However, several contextual factors warrant consideration when interpreting these results. Our sample of agricultural workers maintained consistent nature contact across both intervention and baseline conditions, allowing us to isolate the effects of structured, intentional engagement from those of incidental nature exposure. Moreover, our intervention utilized a standardized healing agricultural program with specific structured activities. Consequently, whether these findings generalize to populations with limited nature contact and whether the mediating role of mindfulness extends to other types of nature-based interventions remain important directions for future research.

6.3. Activities involving higher levels of mindfulness were associated with greater well-being

When comparing the psychological states during participation in all activities throughout the nature-based intervention, both Study 1 and Study 2 indicated that activities fostering a mindfulness state were associated with greater well-being, supporting Hypothesis 2. This result echoes previous research on mindful engagement in natural environments (Macaulay et al., 2022). Macaulay et al. (2022) suggested that deep mindfulness participation in nature enhances individuals' perceptual sensitivity, decentering, and non-reactivity, which may in turn increase the potential benefits people derive from natural settings.

In sum, this research highlights that future designs of nature-based intervention activities should focus on fostering mindful states (related to a focused present and nonjudgmental attitude) to optimize program efficiency. Therefore, it is recommended that future nature-based intervention programs incorporate more mindfulness-guided elements, such as mindful eating (Khan & Zadeh, 2014) or mindful-based art interventions (Coholic & Eys, 2016), which have been shown in previous research to enhance well-being. Actively guiding participants into a state of mindfulness may amplify the positive effects of nature-based interventions on well-being.

6.4. Limitations and future direction

This study has several limitations that warrant consideration. First, the use of experience sampling and day reconstruction methods imposed a significant burden on participants due to frequent data entry requirements. Consequently, each psychological construct was assessed using only a single-item measure. Although this approach is common in such research (Killingsworth & Gilbert, 2010), multi-item measures could enhance the reliability and validity of findings by capturing these constructs more comprehensively. Second, the nature-based interventions in this study took place in domesticated settings, such as farms. The applicability of these findings to wilderness environments warrants further exploration.

Third, the overall response rates were relatively low (40.2 % for Study 1 and 53 % for Study 2). Given the nearly three-month survey duration, participant fatigue likely contributed to this issue—a common challenge in ecological momentary assessment research. One potential solution involves allowing participants to proactively schedule assessments during their available time slots, which has been shown to improve response rates (McLean et al., 2017). Replication studies employing such strategies would strengthen confidence in the present findings.

Fourth, while we found a positive relationship between mindfulness states during nature-based activities and enhanced well-being (Hypothesis 2), this model is correlational, leaving open the possibility of bidirectional influences (Verhulst et al., 2012). Future randomized controlled trials comparing nature-based programs with versus without a mindfulness component are needed to establish causal relationships.

Finally, the current findings are limited to non-clinical populations. Individuals with depression, for instance, exhibit negative attentional bias—a tendency to focus disproportionately on negative information (Mennen et al., 2019). Such cognitive patterns may interfere with the focused, non-judgmental attention required for mindful engagement with natural environments. The mechanisms underlying nature-based interventions in clinical populations merit dedicated investigation.

In conclusion, this study found that mindfulness state serves as the mediating mechanism through which nature-based interventions enhance hedonic and eudaimonic well-being. The study also revealed that, during the nature-based intervention, activities with moments of greater mindfulness were associated with higher levels of well-being. This research not only emphasized the important role that mindfulness states played in nature-based interventions but also suggested that future designs of nature-based intervention programs should specifically incorporate mindfulness elements to maximize the effects of enhancing well-being.

CRediT authorship contribution statement

Peng-Yu Zeng: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Mei-Hsiu Yeh:** Supervision, Resources, Project administration, Funding acquisition. **Su-Ling Yeh:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors have no conflicts of interest to disclose.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvp.2025.102849>.

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